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European Soccer Database Narrative

Modern European soccer generates vast quantities of data across rosters, transfers, matches, and club operations. There are various sources available to access such data, such as ESPN, FotMob, Sofascore, and all league/competition affiliated websites. However, there is no real centralized source for gathering transfers, game statistics, rosters, staff, etc. Even if it is on the same website, it requires clicking through different links and tabs to find the information, which makes gathering it into 1 centralized document or source challenging and intensive. Therefore, many analyses are time intensive or lack a lot of context. This database seeks to unite the information available into 1 unified schema to enable a simple query to gather relevant data and statistics into 1 single source.

This database is designed to support analytics and exploration across both past and current seasons. This opens up a wide array of present-day and longitudinal analyses related to player, team, and manager statistics, club staff operations, and transfer histories. Analysts could easily query the database to gather all of a player's stats for both an individual game, a season, or their entire career, for instance. The data available in this database is intentionally gathered and organized to support the evolving nature of the sport. It encapsulates the complicated world of the global transfer market, the growing amount of stats we can track for a player or team in a match, the evolution of staff roles within an organization, and the history of each individual player, manager, team, or even competition.

In order to account for a large depth of relevant information, the schema tracks not only match-level data but also the relationships between players, teams, competition/leagues, and managers. For instance, we can track the performance of each player involved in a match in the PLAYER_MATCH_STAT table, which records a wide array of statistics related to the player's performance in the match, with room to grow and include more as necessary. We can combine these individual match level analyses with the player's current and former club history, to gather a career or season level glimpse of the player's performance with the PLAYER_TEAM_HISTORY table. We can query the database to acquire information to answer various questions. For example, we can ask "How did a player's performance change after transferring from club A to club B?" or "Who was the top scorer in the team each year?" We can perform similar queries for managers and across clubs as a whole, without having to extract the information from numerous sources, with the MANAGER_TEAM_HISTORY table.

In today’s world, transfers between clubs and the contracts they generate are increasingly important and constantly scrutinized by media and other personnel. With this database, we track transfers quickly and easily with excellent precision. The database keeps track of every transfer between clubs in the database, whether it is a blockbuster permanent deal or a short-term loan. It includes transfer fees, potential bonuses, and the financial promises in the player’s new contract in the TRANSFER and CONTRACT tables. Thus, we can gather and analyze the true financial impact of all transfers. Combined with performance related statistics, we can easily find information to uncover if a big money move was actually worth the money.

Not only can we look at broad league or global level views of the performance, but we can also look within each club at the staff through the STAFF table and its 4 subtypes: COACH, MEDICAL, FRONT_OFFICE, and SCOUT. Each staff member in the organization is assigned to one of these 4 subtypes, excluding the manager, to track who is working behind the scenes at each club. We can look at where clubs are scouting new players, which coaches are creating success in different areas of the game, the qualifications of the medical staff, and who is responsible for the club’s transfers and roster building.

Each match has it’s individual and team-level statistics recorded in the PLAYER_MATCH_STAT, TEAM_MATCH_STAT, and MATCH_EVENT tables. Each match has 2 associated instances in the TEAM_MATCH_STAT table to look at the home and away team’s performance, each player in the match has their own individual entry in the PLAYER_MATCH_STAT table, and each event in the match is recorded in the MATCH_EVENT table like goals, fouls, substitutions, and more, with organizational logic to handle multi-player events. This structure allows us to take a very in depth look at the events and outcome of each match, and track them throughout each season and competition.

Teams in the modern game play in more than just their domestic league. Each club competes in their country’s domestic cup competition(s), and many compete in European-level competitions as well. We can efficiently track a club’s performance across all competitions in all seasons with the STANDING table, which cleverly acts as a bridge between the TEAM, COMPETITION, and SEASON tables. It holds information about the points, wins, and the team’s position in the table or knockout tournament. This allows for querying for the league table (standings) across different seasons, or to get a view of a club’s performance across all competitions.

With these considerations and contexts in mind, the database consists of 20 connected tables: PLAYER, TEAM, MANAGER, STAFF, COACH, MEDICAL, FRONT_OFFICE, SCOUT, MATCH, STADIUM, CONTRACT, TRANSFER, COMPETITION, SEASON, STANDING, PLAYER_MATCH_STAT, TEAM_MATCH_STAT, MATCH_EVENT, PLAYER_TEAM_HISTORY, and MANAGER_TEAM_HISTORY. The structure and rules ensure that analysts and researchers can query, visualize, and organize various accounts of soccer data quickly and easily, whether they want to look at a club or player's performance or monitoring the behind-the-scenes aspects of a club's operations.

Data Dictionary

Table Name	Attribute Name	Contents	Data Type	Format	Range	Required	PK/FK	Reference
PLAYER	PLAYER_ID	Unique ID for players	INT	####...	0-999…	Y	PK	
	PLAYER_FNAME	Player's first name	VARCHAR(30)	Xxxxxx		Y		
	PLAYER_LNAME	Player's last name	VARCHAR(50)	Xxxxxx		Y		
	PLAYER_ALIAS	Player's alias	VARCHAR(30)	Xxxxx		N		
	PLAYER_NATION	Player's nationality	VARCHAR(30)	Xxxxx		Y		
	PLAYER_DOB	Player's Data of Birth	DATE	YYYY-MM-DD	YYYY > 1950	Y		
	PLAYER_POS	Player's primary position	ENUM	XXX	GK, DEF, MID, FWD	Y		
	PLAYER_HEIGHT	Player's height (in)	SMALLINT	##	0-99	Y		
	PLAYER_WEIGHT	Player's Weight (lb)	SMALLINT	###	0-500	Y		
	PLAYER_FOOT	Player's dominant foot	ENUM	X	L, R	Y		
	PLAYER_ACTIVE	Flag for if player is active	BOOL	TRUE, FALSE	TRUE, FALSE	Y		
	CONTRACT_ID	Player's contract IDI	INT	###...	0-999…	N	FK	CONTRACT
TEAM	TEAM_ID	Team's unique ID	INT	###...	0-999…	Y	PK	
	TEAM_NAME	Team's official name	VARCHAR(40)	Xxxx Xxxx		Y		
	TEAM_ABBV	Team's abbreviation	CHAR(3)	XXX		Y		
	TEAM_COUNTRY	Country the team is from	VARCHAR(30)	Xxxx		Y		
	TEAM_CITY	City the team is from	VARCHAR(30)	Xxxx		Y		
	STADIUM_ID	Team's stadium ID	INT	###...	0-999…	Y	FK	STADIUM
	MANAGER_ID	Team's manager ID	INT	###...	0-999	N	FK	MANAGER
MANAGER	MANAGER_ID	Manager's unique ID	INT	###...	0-999…	Y	PK	
	MANAGER_FNAME	Manager's first name	VARCHAR(30)	Xxxx		Y		

	MANAGER_LNAME	Manager's last name	VARCHAR(30)	Xxxx		Y		
	MANAGER_ALIAS	Manager's alias	VARCHAR(30)	Xxxx		N		
	MANAGER_DOB	Manager's date of birth	DATE	YYYY-MM-DD	YYYY > 1900	Y		
	MANAGER_NATION	Manager's nationality	VARCHAR(30)	Xxxx		Y		
	TEAM_ID	Manager's team ID	INT	###...	0-999…	N	FK	MANAGER
	CONTRACT_ID	Manager's Contract ID	INT	###...	0-999…	N	FK	CONTRACT
STAFF	STAFF_ID	Staff member's ID	INT	###...	0-999…	Y	PK	
	STAFF_FNAME	Staff's first name	VARCHAR(30)			Y		
	STAFF_LNAME	Staff's last name	VARCHAR(50)			Y		
	STAFF_TYPE	Staff subtype category	VARCHAR(12)		Coach, Medical, Front Office, Scout	Y		
	TEAM_ID	Staff's team ID	INT	###...		Y	FK	TEAM
COACH	STAFF_ID	Staff's unique ID	INT	###...	0-999…	Y	PK, FK	STAFF
	COACH_TITLE	Coach's title	VARCHAR(50)	Xxxx Xxxx		Y		
	COACH_CERT	Coach's license or credential	VARCHAR(50)	Xxxx Xxxx		Y		
MEDICAL	STAFF_ID	Staff's unique ID	INT	###...	0-999…	Y	PK, FK	STAFF
	MED_TITLE	Medical staff's official title	VARCHAR(50)	Xxxx Xxxx		Y		
	MED_CERT	Medical certification or license	VARCHAR(10)	XXX		Y		
	MED_SPECIAL	Medical specialization	VARCHAR(30)	Xxxx		N		
	MED_CERT_DATE	Date the certification was last awarded or renewed	DATE	YYYY-MM-DD		Y		
FRONT_OFFICE	STAFF_ID	Staff's unique ID	INT	###...	0-999…	Y	PK, FK	STAFF
	FO_TITLE	Front Office member's title	VARCHAR(50)	Xxxx Xxxx		Y		
	FO_DEPT	Front office member's department	VARCHAR(30)	Xxxx Xxxx		Y		
SCOUT	STAFF_ID	Staff's unique ID	INT	###...	0-999…	Y	PK, FK	STAFF
	SCOUT_AREA	Scout's area of scouting	VARCHAR(30)	Xxxx Xxxx		N		
	SCOUT_ACTIVE	Flag for if the scout is currently deployed abroad	BOOLEAN		TRUE, FALSE	Y		
MATCH	MATCH_ID	Match unique ID	INT	###...	0-999…	Y	PK	
	MATCH_DATE	Date of the match	DATE	YYYY-MM-DD		Y		

	MATCH_TIME	Kickoff time of the match	TIME	HH:MM:SS		Y		
	MATCH_ATTENDANCE	Match's official attendance	INT	###...	0-999…	N		
	MATCH_PLAYED	Flag for if the match has been completed	BOOLEAN		TRUE, FALSE	Y		
	COMPETITION_ID	ID for the competition the match is part of	INT	###...	0-999…	Y	FK	COMPETITION
	SEASON_ID	ID for the season the match was played in	INT	###...	0-999…	Y	FK	SEASON
	STADIUM_ID	ID for the stadium of the match	INT	###...	0-999…	Y	FK	STADIUM
STADIUM	STADIUM_ID	Stadium's ID	INT	###	0-999…	Y	PK	
	STADIUM_NAME	Stadium's name	VARCHAR(50)	Xxxx Xxxx		Y		
	STADIUM_CITY	City the stadium is in	VARCHAR(30)	Xxxx		Y		
	STADIUM_COUNTRY	Country the stadium is in	VARCHAR(30)	Xxxx		Y		
	STADIUM_LONG	Longitude coordinate of the stadium	DECIMAL(9,4)	###.######`		Y		
	STADIUM_LAT	Latitude coordinate of the stadium	DECIMAL(9,4)	###.######		Y		
	STADIUM_CAPACITY	Attendance capacity of the stadium	INT	###...	0-999…	Y		
MATCH_EVENT	EVENT_ID	Unique ID for the match event	INT	###...	0-999…	Y	PK	
	EVENT_TYPE	Type of match event	VARCHAR(20)	Xxxxxx		Y		
	EVENT_MINUTE	Minute the event occurred	SMALLINT	###	1-120	Y		
	EVENT_IN_PKS	Flag for if the event occurred in a PK shootout	BOOLEAN	#	TRUE, FALSE	Y		
	EVENT_DESC	Description of the event	TEXT	Xxxx xxxx …		N		
	MATCH_ID	Match ID of the event	INT	###...	0-999…	Y	FK	MATCH
	PLAYER_ID	ID of the player involved	INT	###...	0-999…	Y	FK	PLAYER
	TEAM_ID	ID of the team involved	INT	###...	0-999.	Y	FK	TEAM
TRANSFER	TRAN_ID	ID of the transfer	INT	###...	0-999…	Y	PK	
	TRAN_DATE	Date of the transfer	DATE	YYYY-MM-DD		Y		

	TRAN_AMOUNT	Total dollar amount of transfer	INT	####...	0-999…	Y		
	TRAN_ADDON	Total value of potential add ons	INT	###...	0-999…	N		
	TRAN_TYPE	Type of transfer	VARCHAR(50)	Xxxx		Y		
	TRAN_ISLOAN	Flag for if the transfer is a loan	TINYINT	#	0, 1	Y		
	PLAYER_ID	ID of the player involved	INT	###...	0-999…	Y	FK	PLAYER
	FROM_TEAM_ID	ID of the player's former team	INT	###...	0-999…	N	FK	TEAM
	TO_TEAM_ID	ID of the player's new team	INT	###...	0-999…	Y	FK	TEAM
	CONTRACT_ID	Player's new contract ID	INT	###...	0-999…	Y	FK	CONTRACT
	SEASON_ID	ID of the season for this transfer	INT	###...	0-999..	Y	FK	SEASON
COMPETITION	COMP_ID	ID of the competition	INT	###...	0-999.	Y	PK	
	COMP_NAME	Name of the competition	VARCHAR(50)	Xxxx Xxxx		Y		
	COMP_COUNTRY	Country of the compeition	VARCHAR(30)	Xxxx		N		
	COMP_TYPE	Format of the competition	ENUM	Xxxx	League, Knockout, Hybrid	Y		
	COMP_TIER	Tier of the competition	TINYINT	#	1-4	N		
	COMP_GOVERN	Governing body of the competition	VARCHAR(10)	XXXX		Y		
CONTRACT	CONTRACT_ID	ID of the contract	INT	###...	0-999…	Y	PK	
	CONTRACT_START	Start date of the contract	DATE	YYYY-MM-DD		Y		
	CONTRACT_END	End date of the contract	DATE	YYYY-MM-DD	CONTRACT_END > CONTRACT_START	Y		
	CONTRACT_SALARY	Person's contracted salary in dollars per week	INT	###...	1-999…	Y		
	CONTRACT_BONUS	Total potential bonuses	INT	###...	1-999…	N		
	CONTRACT_RELEASE	Value of contract's release clause	INT	###...	1-999…	N		
	CONTRACT_ACTIVE	Flag for if the contract is active	BOOLEAN	#	0, 1	Y		

	CONTRACT_TYPE	Type of the contract	ENUM	Xxxx	Manager or Player	Y		
	PLAYER_ID	ID of the player in the contract	INT	###...	0-999…	N	FK	PLAYER
	MANAGER_ID	ID of the manager in the contract	INT	###...	0-999…	N	FK	MANAGER
	TEAM_ID	ID of the contract's team	INT	###...	0-999…	Y	FK	TEAM
	TRAN_ID	ID of the transfer associated with the contract	INT	###...	0-999…	N	FK	TRANSFER
STANDING	SEASON_ID	ID of the season involved	INT	###...	0-999…	Y	PK, FK	SEASON
	COMPETITION_ID	ID of the competition for this standings record	INT	###...	0-999…	Y	PK, FK	COMPETITION
	TEAM_ID	ID of the team for this record	INT	###...	0-999…	Y	PK, FK	TEAM
	STAND_GAMES	# of games played	SMALLINT	## or #	0-50	Y		
	STAND_WINS	# of wins	SMALLINT	## or #	0-STAND_GAMES	Y		
	STAND_DRAWS	# of ties/draws	SMALLINT	## or #	0-STAND_GAMES	Y		
	STAND_LOSSES	# of losses	SMALLINT	## or #	0-STAND_GAMES	Y		
	STAND_POINTS	# of points	SMALLINT	# or ## or ###	-50 – 3*STAND_GAMES	Y		
	STAND_GS	# of goals scored	SMALLINT	# or ## or ###	0-999	Y		
	STAND_GA	# of goals against	SMALLINT	# or ## or ###	0-999	Y		
	STAND_GD	Goal differential	SMALLINT	###	-999 to 999	Y		
	STAND_POS	Position in the standings	SMALLINT	# or ##	1 – 50	Y		
	STAND_DATE	Timestamp of last standings update	DATE	YYYY-MM-DD		Y		
PLAYER_MATCH_STAT	PLAYER_ID	ID for the player involved	INT	###...	0-999…	Y	PK/FK	PLAYER
	MATCH_ID	ID for the match involved	INT	###...	0-999…	Y	PK/FK	MATCH
	PM_MINUTES	Minutes the player played	SMALLINT	###...	0-120	Y		
	PM_GOALS	# of goals scored	SMALLINT	##	0-50	Y		
	PM_ASSISTS	# of assists	SMALLINT	##	0-50	Y		
	PM_SHOTS	# of shots taken	SMALLINT	##	0-99	Y		
	PM_SOG	# of shots on goal	SMALLINT	##	0-PM_SHOTS	Y		
	PM_PKS	# of PKs scored	SMALLINT	##	0-PM_GOALS	Y		
	PM_PKM	# of PKs missed	SMALLINT	##	0-PM_SHOTS	Y		

	PM_PA	# of passes attempted	SMALLINT	###	0-400	Y		
	PM_PC	# of passes completed	SMALLINT	###	0-400	Y		
	PM_TACKLES	# of tackles completed	SMALLINT	##	0-50	Y		
	PM_FOULS_C	# of fouls committed	SMALLINT	##	0-20	Y		
	PM_FOULS_A	# of times fouled	SMALLINT	##	0-20	Y		
	PM_YC	# of yellow cards	SMALLINT	#	0-2	Y		
	PM_RC	# of red cards	SMALLINT	#	0-1	Y		
	PM_INTERCEPT	# of interceptions	SMALLINT	##	0-50	Y		
	PM_SAVES	# of saves made	SMALLINT	##	0-50	Y		
	PM_CONCEDE	# of goals conceded	SMALLINT	##	0-50	Y		
	PM_PK_SAVE	# of PKs saved	SMALLINT	##	0-PM_SAVES	y		
	PM_START	Flag for if the player was a starter	BOOLEAN	#	0 or 1	Y		
TEAM_MATCH_STAT	TEAM_ID	ID of the team involved	INT	###...	0-999…	Y	PK/FK	TEAM
	MATCH_ID	ID of the match involved	INT	###...	0-999…	Y	PK/FK	MATCH
	TM_HOME	Indicator if the team was at home	BOOLEAN	#	0-1	Y		
	TM_GS	# of goals scored	SMALLINT	##	0-50	Y		
	TM_GA	# of goals conceded	SMALLINT	##	0-50	Y		
	TM_POSS	% possession this team had	DECIMAL(3,1)	##	00.1-99.9	Y		
	TM_SHOTS	# of shots taken	SMALLINT	##	0-99	Y		
	TM_SOG	# of shots on target	SMALLINT	##	0-TM_SHOTS	Y		
	TM_FOULS_C	# of fouls conceded	SMALLINT	##	0-50	Y		
	TM_FOULS_A	# of fouls won	SMALLINT	##	0-50	Y		
	TM_YC	# of yellow cards	SMALLINT	##	0-40	Y		
	TM_RC	# of red cards	SMALLINT	##	0-20	Y		
	TM_PA	# of passes attempted	SMALLINT	###	0-1000	Y		
	TM_PC	# of passes completed	SMALLINT	###	0-TM_PA	Y		
	TM_CORNER	# of corner kicks	SMALLINT	##	0-50	Y		
	TM_RESULT	String for the outcome	VARCHAR(4)	Xxxx	Win, Loss, or Draw	Y		
SEASON	SEASON_ID	ID of the season	INT	###...	0-999…	Y	PK	
	SEASON_START	Start date of the season	DATE	YYYY-MM-DD		Y		

	SEASON_END	End date of the season	DATE	YYYY-MM-DD		Y		
	SEASON_ACTIVE	Flag for if the season is active	BOOLEAN	#	0-1	Y		
PLAYER_TEAM_HISTORY	PLAYER_ID	ID of the player	INT	###...	0-999…	Y	PK/FK	PLAYER
	TEAM_ID	ID of the team	INT	###...	0-999…	Y	PK/FK	TEAM
	PT_START	Date the player joined the team	DATE	YYYY-MM-DD		Y	PK	
	PT_END	Date the player left the team	DATE	YYYY-MM-DD	> PT_START	N		
	PT_JERSEY	Jersey number of the player at the team	SMALLINT	# or ##	1-99	Y		
MANAGER_TEAM_HISTORY	MANAGER_ID	ID of the manager	INT	###...	0-999…	Y	PK/FK	MANAGER
	TEAM_ID	ID of the team	INT	###...	0-999…	Y	PK/FK	TEAM
	MT_START	Date the manager joined the team	DATE	YYYY-MM-DD		Y	PK	
	MT_END	Date the manager left the team	DATE	YYYY-MM-DD		N		

Entity Relationship Model

ENTITY	**RELATIONSHIP**	**CONNECTIVITY**	**ENTITY**
PLAYER	has	(1:M)	PLAYER_TEAM_HISTORY
PLAYER	completes	(0:M)	TRANSFER
PLAYER	completes	(0:M)	MATCH_EVENT
PLAYER	completes	(0:M)	PLAYER_MATCH_STAT
PLAYER	has signed	(0:1)	CONTRACT
TEAM	has	(1:M)	PLAYER_TEAM_HISTORY

TEAM	employs	(1:M)	STAFF
TEAM	has	(1:M)	STANDING
TEAM	completes	(0:M)	TRANSFER
TEAM	owns	(1:M)	CONTRACT
TEAM	has	(1:M)	MANAGER_TEAM_HISTORY
TEAM	is managed by	(0:1)	MANAGER
TEAM	completes	(0,M)	MATCH_EVENT
MANAGER	coaches	(0:1)	TEAM
MANAGER	has	(1:M)	MANAGER_TEAM_HISTORY
MANAGER	signs	(1:1)	CONTRACT
STAFF	specializes in	(0:1) (subtype)	COACH
STAFF	specializes in	(0:1) (subtype)	MEDICAL
STAFF	specializes in	(0:1) (subtype)	FRONT_OFFICE
STAFF	specializes in	(0:1) (subtype)	SCOUT
MATCH	contains	(1:M)	TEAM_MATCH_STAT
MATCH	contains	(0:M)	MATCH_EVENT
MATCH	has	(1:M)	PLAYER_MATCH_STAT

STADIUM	hosts	(0:M)	MATCH
STADIUM	is associated with	(0:1)	TEAM
COMPETITION	has	(1:M)	MATCH
COMPETITION	has	(1:M)	STANDING
SEASON	has	(0:M)	TRANSFER
SEASON	holds	(1:M)	STANDING
SEASON	holds	(1:M)	MATCH
CONTRACT	references	(0:1)	TRANSFER

Business Rules:

PLAYER/PLAYER_TEAM_HISTORY

1. Each PLAYER can have 1 or many PLAYER_TEAM_HISTORY records
2. Each PLAYER_TEAM_HISTORY record refers to exactly 1 player
3. PLAYER_TEAM_HISTORY.PT_START must not be null, as it indicates when the player started playing on this team.
4. PLAYER_TEAM_HISTORY.PT_END can be null, but only if the PLAYER is currently on the associated TEAM.
5. A PLAYER_TEAM_HISTORY.PT_START record cannot overlap with a PLAYER_TEAM_HISTORY.PT_END record for the same PLAYER_ID.

PLAYER/TRANSFER

1. Each PLAYER completes 0 or many TRANSFERs.
2. Each TRANSFER is associated with exactly 1 PLAYER.

PLAYER/MATCH_EVENT

1. Each PLAYER may complete 0 or many MATCH_EVENT records.
2. Each MATCH_EVENT record refers to exactly 1 PLAYER

PLAYER/PLAYER_MATCH_STAT

1. Each PLAYER can be involved in 0 or many PLAYER_MATCH_STAT records
2. Each PLAYER_MATCH_STAT record belongs to exactly 1 player
3. If PM_START = 1, then PM_MINUTES > 0 (the player started the match).

PLAYER/CONTRACT

1. Each PLAYER may have 0 or 1 CONTRACT records
2. Each CONTRACT is associated with exactly 1 PLAYER
3. CONTRACT.CONTRACT_TYPE must = 'Player' and CONTRACT.PERSON_ID must match a valid PLAYER.PLAYER_ID for a valid PLAYER/CONTRACT relationship to exist.
4. CONTRACT_END must be after CONTRACT_START

TEAM/PLAYER_TEAM_HISTORY

1. Each TEAM is associated with 1 or many PLAYER_TEAM_HISTORY records. A team must have at least 1, otherwise it has never had a roster and will not be included.
2. Each PLAYER_TEAM_HISTORY record is related to exactly 1 TEAM.

TEAM/STAFF

1. Each TEAM employs 1 or many STAFF members.
2. Each STAFF member is employed by exactly 1 TEAM.

TEAM/STANDING

1. Each TEAM participates in 1 or many STANDING records.
2. Each STANDING record is associated with exactly 1 TEAM.
3. STAND_GAMES = STAND_WINS + STAND_DRAWS + STAND_LOSS, because a TEAM must not have more wins/draws/losses than games played or vice versa.
4. STAND_POINTS cannot exceed 3*STAND_WINS + 3*STAND_DRAWS. It can be lower, but this would indicate the TEAM had points removed for disciplinary reasons.
5. For different TEAMs associated with the same COMPETITION and SEASON in the STANDING TABLE (different STANDING.TEAM_ID, same COMP_ID and SEASON_ID), the ordering of STAND_POS must match the ordering of STAND_POINTS
   a. TEAM_ID with higher STAND_POINTS value must have a lower value for STAND_POS

TEAM/TRANSFER

1. Each TEAM can complete 0 or many TRANSFER records either as the origin or destination.
2. Each TRANSFER must reference a from TEAM (TRAN.FROM_TEAM_ID) and a to TEAM (TRAN.TO_TEAM_ID).
3. The “from” TEAM and “to” TEAM must not be the same. Thus, TRAN.FROM_TEAM_ID cannot be the same as TRAN.TO_TEAM_ID.

TEAM/CONTRACT

1. Each TEAM owns many CONTRACT records.
2. Each CONTRACT belongs to exactly 1 TEAM.
3. Every TEAM must cannot have more than 1 CONTRACT associated with a manager. This means that the count of CONTRACT items where CONTRACT.CONTRACT_TYPE = ‘Manager’ associated with a TEAM must be 0 or 1.

TEAM/MANAGER_TEAM_HISTORY

1. Each TEAM must have at least 1 MANAGER_TEAM_HISTORY record.
2. Each MANAGER_TEAM_HISTORY instance is related to exactly 1 MANAGER.

TEAM/MATCH_EVENT

1. Each TEAM can complete 0 or many MATCH_EVENT records.
2. Each MATCH_EVENT is completed by exactly 1 TEAM.
3. In the case where a MATCH_EVENT involves players from 2 different TEAM instances (e.g. a foul), 2 MATCH_EVENT records will be recorded, each linking to a different TEAM record.

TEAM/MANAGER

1. Every TEAM is coached by 0 or 1 MANAGER.
2. Every MANAGER coaches 0 or 1 TEAM.
3. TEAM.MANAGER_ID is set to NULL when a TEAM fires a MANAGER, or a MANAGER’s contract expires and a new MANAGER or an interim MANAGER has not yet been hired.

MANAGER/TEAM

1. Every MANAGER coaches 0 or 1 TEAM.
2. Every TEAM is coached by exactly 1 MANAGER.
3. If MANAGER.TEAM_ID is NULL, the MANAGER is not currently employed by a TEAM. Thus, MANAGER.CONTRACT_ID must be NULL.

MANAGER/MANAGER_TEAM_HISTORY

1. Each MANAGER must have 1 or many MANAGER_TEAM_HISTORY records.
2. Each MANAGER_TEAM_HISTORY record is related to exactly 1 MANAGER.
3. MT_START must be before MT_END, unless MT_END is NULL, which implies the MANAGER is currently managing that TEAM. MT_START cannot be NULL.

MANAGER/CONTRACT

1. Each MANAGER can sign 0 or 1 CONTRACT.
2. Each CONTRACT is related to exactly 1 MANAGER.
3. CONTRACT.CONTRACT_TYPE must = 'Manager' and CONTRACT.PERSON_ID must match a valid MANAGER.MANAGER_ID for a valid relationship to exist.

STAFF/COACH

1. COACH is a disjoint, partial subtype of the STAFF supertype. Not every STAFF member is a COACH, and no STAFF member can be a COACH and another subtype member.
2. Every COACH has a matching STAFF supertype instance.
3. STAFF.STAFF_TYPE must = 'Coach' for the STAFF member to be a COACH.

STAFF/MEDICAL

1. MEDICAL is a disjoint, partial subtype of the STAFF supertype. Thus, not every STAFF member is part of the MEDICAL team, and no STAFF member can be on the MEDICAL team and be another subtype instance.
2. Each MEDICAL instance has a matching STAFF supertype instance.
3. STAFF.STAFF_TYPE must = 'Medical' for the STAFF member to be on the MEDICAL team.

STAFF/FRONT_OFFICE

1. FRONT_OFFICE is a disjoint, partial subtype of the STAFF supertype. Not every STAFF member is in the FRONT_OFFICE, and no STAFF member can be in the FRONT_OFFICE and another subtype.

2. Every FRONT_OFFICE member has a matching STAFF supertype instance.
3. STAFF.STAFF_TYPE must = ‘FRONT_OFFICE’ for the STAFF member to be in the FRONT_OFFICE.

STAFF/SCOUT

1. SCOUT is a disjoint, partial subtype of the STAFF supertype. Not every STAFF member is a SCOUT, and no STAFF member can be a SCOUT and another subtype member.
2. Every SCOUT has a matching STAFF supertype instance.
3. STAFF.STAFF_TYPE must = ‘SCOUT’ for the STAFF member to be a SCOUT.
4. If SCOUT.ACTIVE = 1, then the scout is currently deployed abroad to their SCOUT_AREA.

MATCH/TEAM_MATCH_STAT

1. Each MATCH has exactly 2 TEAM_MATCH_STAT records. 1 is for the home team and the other is for the away team.
    a. Thus, for TEAM_MATCH_STAT instances with the same TEAM_MATCH_STAT.MATCH_ID, one must have TM_HOME = 1 and the other must have TM_HOME = 0.
2. Each TEAM_MATCH_STAT record is associated with exactly 1 MATCH.

MATCH/MATCH_EVENT

1. Each MATCH can contain 0 or many MATCH_EVENT records. It is theoretically possible for a MATCH to have no MATCH_EVENT records occur, since kickoff, halftime, and full time are not recorded, so it is possible to have 0 (though unlikely).
2. Each MATCH_EVENT must be associated with exactly 1 MATCH.
3. For any MATCH_EVENT instance that involves 2 or more players (e.g. substitution, foul, shot with a save), 2 MATCH_EVENT instances will be added, 1 for each player.
    a. MATCH_EVENT.EVENT_TYPE = ‘Substitution in’ and ‘Substitution out’ for the 2 separate players involved in the substitution
    b. EVENT_TYPE = ‘Foul’ or ‘Was Fouled’ for the player committing/winning the foul
    c. EVENT_TYPE = ‘Shot’ or ‘Save’ for a shot with a save.

MATCH/PLAYER_MATCH_STAT

1. Each match contains at least 22 PLAYER_MATCH_STAT records. This includes the stats for all 22 starters on both teams, and any substitutes who make an appearance.

2. Each PLAYER_MATCH_STAT instance is associated with exactly 1 MATCH.

STADIUM/MATCH

1. Each STADIUM can host 0 or many MATCH instances.
2. Each MATCH is played in exactly 1 STADIUM.

STADIUM/TEAM

1. Each STADIUM plays host to 0 or 1 TEAM.
2. Each TEAM plays in exactly 1 home STADIUM.

COMPETITION/MATCH

1. Every COMPETITION contains 1 or many MATCH records.
2. Every MATCH is associated with exactly 1 COMPETITION.

COMPETITION/STANDING

1. Each COMPETITION contains at least 1 STANDING record.
2. Each STANDING record is associated with exactly 1 COMPETITION.

SEASON/TRANSFER

1. Every SEASON can contain 0 or many TRANSFER instances.
2. Each TRANSFER must be associated with exactly 1 SEASON.
3. TRANSFER.TRAN_DATE must be between SEASON.SEASON_START and SEASON.SEASON_END (or if SEASON.SEASON_END is NULL, TRANSFER.TRAN_DATE must be after SEASON.SEASON_START for transfers in the active SEASON)

SEASON/STANDING

1. Every SEASON contains 1 or many STANDING records.
2. Each STANDING is associated with exactly 1 SEASON.

SEASON/MATCH

1. Every SEASON contains 1 or more MATCH records.
2. Each MATCH is associated with exactly 1 SEASON.

3. MATCH.MATCH_DATE must be after the associated SEASON.SEASON_START date and before the SEASON.SEASON_END date if the MATCH occurred in a past (inactive) season.

CONTRACT/TRANSFER

1. Each CONTRACT may reference 0 or 1 TRANSFER.
2. Each TRANSFER must reference exactly 1 CONTRACT.
3. CONTRACT.CONTRACT_START must be on or after TRANSFER.TRAN_DATE.
4. If CONTRACT.CONTRACT_TYPE = ‘Manager’, the CONTRACT cannot reference a TRANSFER.
5. CONTRACT.TEAM_ID must match TRANSFER.TO_TEAM_ID if the CONTRACT is associated with a TRANSFER.

<u>Intended Use</u>

Ideally, this database would be used by analysts, either within a club or on their own, to support and speed up data collection for analysis. It will be dynamically updated, both with manual entry and code for data extraction to help ensure an up-to-date, centralized view of soccer statistics and information. Multiple views and indices will be created to speed up common queries, such as a specific player/club’s statistics over a season, competition, individual match, or career. Thus, the user will just be able to write a simple query and the database will combine and parse through multiple tables to extract and combine all relevant information.

The MATCH_EVENT table will allow for automatic updates to the PLAYER_MATCH_STAT and TEAM_MATCH_STAT tables. Since the MATCH_EVENT table requires a linked player and team, each time an event is added to the system, it will automatically increment the associated player or team’s attributes for that match in the associated match stat tables. Thus, the designer does not have to individually add information about each event multiple times. Further, with the STANDING table, each time a match is concluded, the results will be input into MATCH and its related tables, which will automatically increment the points total (for league matches), and alter the STAND_POS value for each team in the associated competition. Thus, the designer does not have to individually change the standings after the conclusion of each match.

Similarly, transfers and contracts will automatically update the roster through the PLAYER_TEAM_HISTORY table. If a new transfer or contract is initiated, the date the player left their old club, the start of their time at their new club, and the proper IDs will be altered to ensure the roster’s are updated quickly and easily. Again, this automation allows for simpler use for the designer and user, ensuring information is accurate and consistent across the database.

Relational Schema

PLAYER(**PLAYER_ID**, PLAYER_FNAME, PLAYER_LNAME, PLAYER_ALIAS, PLAYER_NATION, PLAYER_DOB, PLAYER_POS, PLAYER_HEIGHT, PLAYER_WEIGHT, PLAYER_FOOT, PLAYER_ACTIVE, CONTRACT_ID)

TEAM(**TEAM_ID**, TEAM_NAME, TEAM_ABBV, TEAM_COUNTRY, TEAM_CITY, STADIUM_ID, MANAGER_ID)

MANAGER(**MANAGER_ID**, MANAGER_FNAME, MANAGER_LNAME, MANAGER_ALIAS, MANAGER_DOB, MANAGER_NATION, TEAM_ID, CONTRACT_ID)

STAFF(**STAFF_ID**, STAFF_FNAME, STAFF_LNAME, STAFF_TYPE, TEAM_ID)

COACH(**STAFF_ID**, COACH_TITLE, COACH_CERT)

MEDICAL(**STAFF_ID**, MED_TITLE, MED_CERT, MED_SPECIAL, MED_CERT_DATE)

FRONT_OFFICE(**STAFF_ID**, FO_TITLE, FO_DEPT)

SCOUT(**STAFF_ID**, SCOUNT_AREA, SCOUT_ACTIVE)

MATCH(**MATCH_ID**, MATCH_DATE, MATCH_TIME, MATCH_ATTENDANCE, MATCH_PLAYED, COMPETITION_ID, SEASON_ID, STADIUM_ID)

STADIUM(**STADIUM_ID**, STADIUM_NAME, STADIUM_CITY, STADIUM_COUNTRY, STADIUM_LONG, STADIUM_LAT, STADIUM_CAPACITY)

MATCH_EVENT(**EVENT_ID**, EVENT_TYPE, EVENT_MINUTE, EVENT_IN_PKS, EVENT_DESC, MATCH_ID, PLAYER_ID, TEAM_ID)

TRANSFER(**TRAN_ID**, TRAN_DATE, TRAN_AMOUNT, TRAN_ADDON, TRAN_TYPE, TRAN_ISLOAN, PLAYER_ID, FROM_TEAM_ID, TO_TEAM_ID, CONTRACT_ID, SEASON_ID)

COMPETITION(**COMP_ID**, COMP_NAME, COMP_COUNTRY, COMP_TYPE, COMP_TIER, COMP_GOVERN)

CONTRACT(**CONTRACT_ID**, CONTRACT_START, CONTRACT_END, CONTRACT_SALARY, CONTRACT_BONUS, CONTRACT_RELEASE, CONTRACT_ACTIVE, CONTRACT_TYPE, PLAYER_ID, MANAGER_ID, TEAM_ID, TRAN_ID)

STANDING(**SEASON_ID, COMPETITION_ID, TEAM_ID**, STAND_GAMES, STAND_WINS, STAND_DRAWS, STAND_LOSSES, STAND_POINTS, STAND_GS, STAND_GA, STAND_GD, STAND_POS, STAND_DATE)

PLAYER_MATCH_STAT(**PLAYER_ID, MATCH_ID**, PM_MINUTES, PM_GOALS, PM_ASSISTS, PM_SHOTS, PM_SOG, PM_PKS, PM_PKM, PM_PA, PM_PC, PM_TACKLES, PM_FOULS_C, PM_FOULS_A, PM_YC, PM_RC, PM_INTERCEPT, PM_SAVES, PM_CONCEDE, PM_PK_SAVE, PM_START)

TEAM_MATCH_STAT(**TEAM_ID, MATCH_ID**, TM_HOME, TM_GS, TM_GA, TM_POSS, TM_SHOTS, TM_SOG, TM_FOULS_C, TM_FOULS_A, TM_YC, TM_RC, TM_PA, TM_PC, TM_CORNER, TM_RESULT)

SEASON(**SEASON_ID**, SEASON_START, SEASON_END, SEASON_ACTIVE)

PLAYER_TEAM_HISTORY(**PLAYER_ID, TEAM_ID, PT_START**, PT_END, PT_JERSEY)

MANAGER_TEAM_HISTORY(**MANAGER_ID, TEAM_ID, MT_START**, MT_END)

Table Normalization

Pre Normalization

This table is used to record all players as well as their current and former teams. Each row represents a player at a certain team at a certain time.

PLAYER_ID	PLAYER_NAME	PLAYER_ALIAS	PLAYER_NATION	PLAYER_DOB	PLAYER_HEIGHT	PLAYER_WEIGHT	PLAYER_FOOT	PLAYER_ACTIVE	CONTRACT_ID	TEAM_NAME	TEAM_ABBV	TEAM_COUNTRY	TEAM_CITY	STADIUM_ID	MANAGER_ID	PT_START	PT_END	PT_JERSEY
1	Marcus Rashford	M. Rashford	England	1997-10-31	73	176	R	Y	1	Manchester United	MU	England	Manchester	1	1	2016-02-25	2025-01-01	10
									2	Aston Villa	AVL	England	Birmingham	2	2	2025-01-01	2025-05-31	9
									3	Barcelona	BAR	Spain	Barcelona	3	3	2025-07-01		14
2	Antony Matheus dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	4	Sao Paulo	SAP	Brazil	Sao Paulo	4	4	2018-11-15	2020-02-23	11
									5	Ajax	AJX	Netherlands	Amsterdam	5	5	2020-02-23	2022-08-30	39
									6	Manchester United	MU	England	Manchester	1	5	2022-08-30	2025-01-25	21
									7	Real Betis	BET	Spain	Seville	6	6	2025-01-25	2025-06-30	7
									8							2025-09-01		

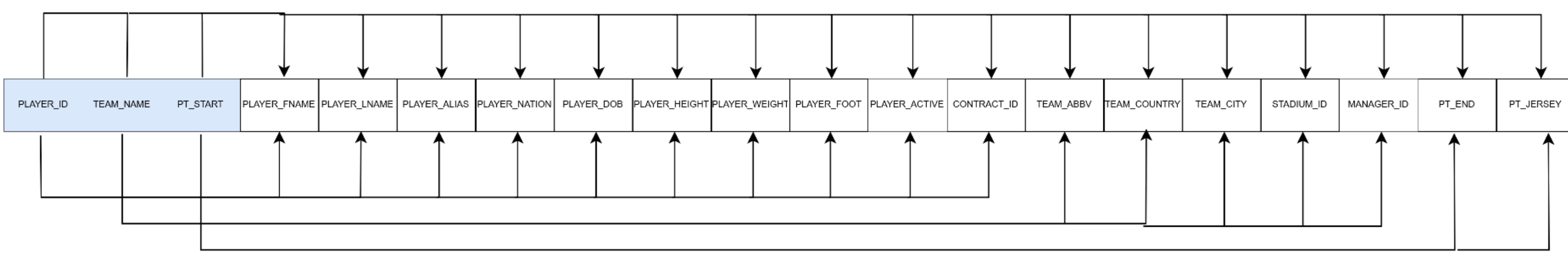
1st Normal Form

In the de-normalized form, we have multiple empty values that cannot be empty, and the PLAYER_NAME attribute is non-atomic. We need to split PLAYER_NAME into PLAYER_FNAME and PLAYER_LNAME. We can also identify a composite primary key consisting of PLAYER_ID, TEAM_NAME, and PT_START. This identifies a row as a player's stint with a specific team. However, there are still many partial dependencies. PLAYER_ID determines PLAYER_FNAME, PLAYER_LNAME, PLAYER_ALIAS, PLAYER_NATION, PLAYER_DOB, PLAYER_HEIGHT, PLAYER_WEIGHT, PLAYER_FOOT, PLAYER_ACTIVE, and CONTRACT_ID. TEAM_NAME determines

TEAM_ABBV, TEAM_COUNTRY, TEAM_CITY, STADIUM_ID, and MANAGER_ID. Lastly, PT_START determines PT_END and PT_JERSEY.

PLAYER_ID	PLAYER_FNAME	PLAYER_LNAME	PLAYER_ALIAS	PLAYER_NATION	PLAYER_DOB	PLAYER_HEIGHT	PLAYER_WEIGHT	PLAYER_FOOT	PLAYER_ACTIVE	CONTRACT_ID	TEAM_NAME	TEAM_ABBV	TEAM_COUNTRY	TEAM_CITY	STADIUM_ID	MANAGER_ID	PT_START	PT_END	PT_JERSEY
1	Marcus	Rashford		England	1997-10-31	73	176	R	Y	1	Manchester United	MU	England	Manchester	1	1	2016-02-25	2025-01-01	10
1	Marcus	Rashford		England	1997-10-31	73	176	R	Y	2	Aston Villa	AVL	England	Birmingham	2	2	2025-01-01	2025-05-31	9
1	Marcus	Rashford		England	1997-10-31	73	176	R	Y	3	Barcelona	BAR	Spain	Barcelona	3	3	2025-07-01		14
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	4	Sao Paulo	SAP	Brazil	Sao Paulo	4	4	2018-11-15	2020-02-23	11
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	5	Ajax	AJX	Netherlands	Amsterdam	5	5	2020-02-23	2022-08-30	39
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	6	Manchester United	MU	England	Manchester	1	5	2022-08-30	2025-01-25	21
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	7	Real Betis	BET	Spain	Seville	6	6	2025-01-25	2025-06-30	7
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	8	Real Betis	BET	Spain	Seville	6	6	2025-09-01		7

1st Normal Form Dependency Diagram



1NF(**PLAYER_ID, TEAM_NAME, PT_START**, PLAYER_FNAME, PLAYER_LNAME, PLAYER_ALIAS, PLAYER_NATION, PLAYER_DOB, PLAYER_HEIGHT, PLAYER_WEIGHT, PLAYER_FOOT, PLAYER_ACTIVE, CONTRACT_ID, TEAM_ABBV, TEAM_COUNTRY, TEAM_CITY, STADIUM_ID, MANAGER_ID, PT_END, PT_JERSEY)

Partial Dependencies

(PLAYER_ID → PLAYER_FNAME, PLAYER_LNAME, PLAYER_ALIAS, PLAYER_NATION, PLAYER_DOB, PLAYER_HEIGHT, PLAYER_WEIGHT, PLAYER_FOOT, PLAYER_ACTIVE, CONTRACT_ID)

(TEAM_NAME → TEAM_ABBV, TEAM_COUNTRY, TEAM_CITY, STADIUM_ID, MANAGER_ID)

(PT_START → PT_END, PT_JERSEY)

2nd Normal Form

Since we had 3 partial dependencies in our 1st normal form, we break these into separate tables. However, we notice from the table that a team can have many players, and a player can have many teams throughout their career. Thus, when assigning foreign keys, we can actually use the table built from PT_START as a bridge between the PLAYER_ID and TEAM_NAME dependent tables. Also, to use better, guaranteed unique primary keys, the primary key for the TEAM table will be changed to TEAM_ID. Then, the PLAYER-TEAM bridge table will have a primary key of (PLAYER_ID, TEAM_ID, PT_START). This way, we can easily access the player/team relationship for that stint at the club.

PLAYER

PLAYER_ID	PLAYER_FNAME	PLAYER_LNAME	PLAYER_ALIAS	PLAYER_NATION	PLAYER_DOB	PLAYER_HEIGHT	PLAYER_WEIGHT	PLAYER_FOOT	PLAYER_ACTIVE	CONTRACT_ID
1	Marcus	Rashford		England	1997-10-31	73	176	R	Y	3
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	8

TEAM

TEAM_ID	TEAM_NAME	TEAM_ABBV	TEAM_COUNTRY	TEAM_CITY	STADIUM_ID	MANAGER_ID
1	Manchester United	MU	England	Manchester	1	1
2	Aston Villa	AVL	England	Birmingham	2	2
3	Barcelona	BAR	Spain	Barcelona	3	3
4	Sao Paulo	SAP	Brazil	Sao Paulo	4	4
5	Ajax	AJX	Netherlands	Amsterdam	5	5
6	Real Betis	BET	Spain	Seville	6	6

PLAYER_TEAM_HISTORY

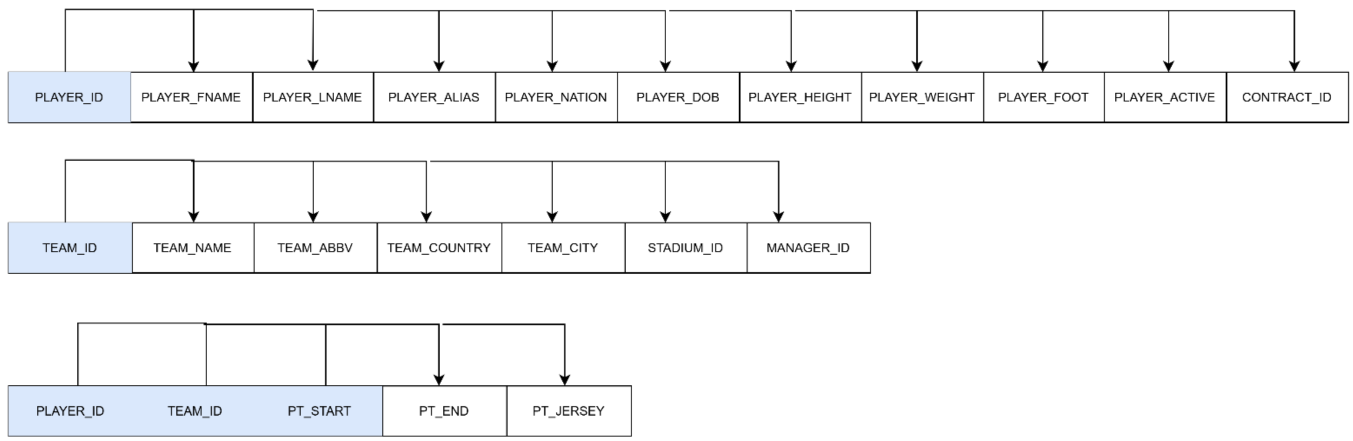
PLAYER_ID	TEAM_ID	PT_START	PT_END	PT_JERSEY
1	1	2016-02-25	2025-01-01	10
1	2	2025-01-01	2025-05-31	9
1	3	2025-07-01		14
2	4	2018-11-15	2020-02-23	11
2	5	2020-02-23	2022-08-30	39
2	1	2022-08-30	2025-01-25	21
2	6	2025-01-25	2025-06-30	7
2	6	2025-09-01		7

2nd Normal Form Dependency Diagram

PLAYER(**PLAYER_ID**, PLAYER_FNAME, PLAYER_LNAME, PLAYER_ALIAS, PLAYER_NATION, PLAYER_DOB, PLAYER_HEIGHT, PLAYER_WEIGHT, PLAYER_FOOT, PLAYER_ACTIVE, CONTRACT_ID)

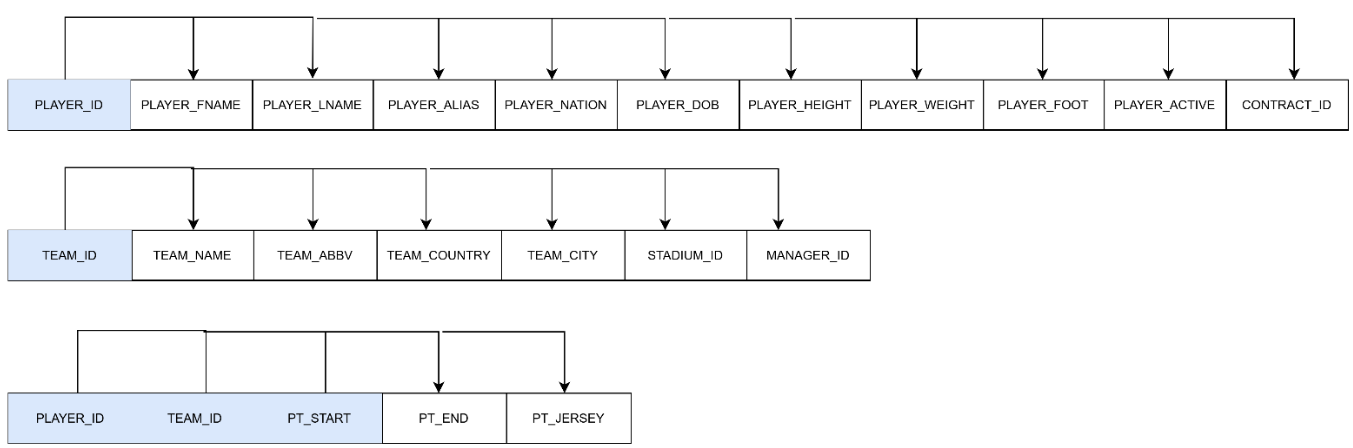
TEAM(**TEAM_ID**, TEAM_NAME, TEAM_ABBV, TEAM_COUNTRY, TEAM_CITY, STADIUM_ID, MANAGER_ID)

PLAYER_TEAM_HISTORY(**PLAYER_ID, TEAM_ID, PT_START**, PT_END, PT_JERSEY)



3^{RD} Normal Form

Since we had no transitive dependencies in 1^{st} normal form or 2^{nd} normal form, the tables are now in 3^{rd} normal form as well. Thus, no changes need to be made to the schema.



PLAYER

PLAYER_ID	PLAYER_FNAME	PLAYER_LNAME	PLAYER_ALIAS	PLAYER_NATION	PLAYER_DOB	PLAYER_HEIGHT	PLAYER_WEIGHT	PLAYER_FOOT	PLAYER_ACTIVE	CONTRACT_ID
1	Marcus	Rashford		England	1997-10-31	73	176	R	Y	3
2	Antony Matheus	dos Santos	Antony	Brazil	2000-02-24	69	139	L	Y	8

TEAM

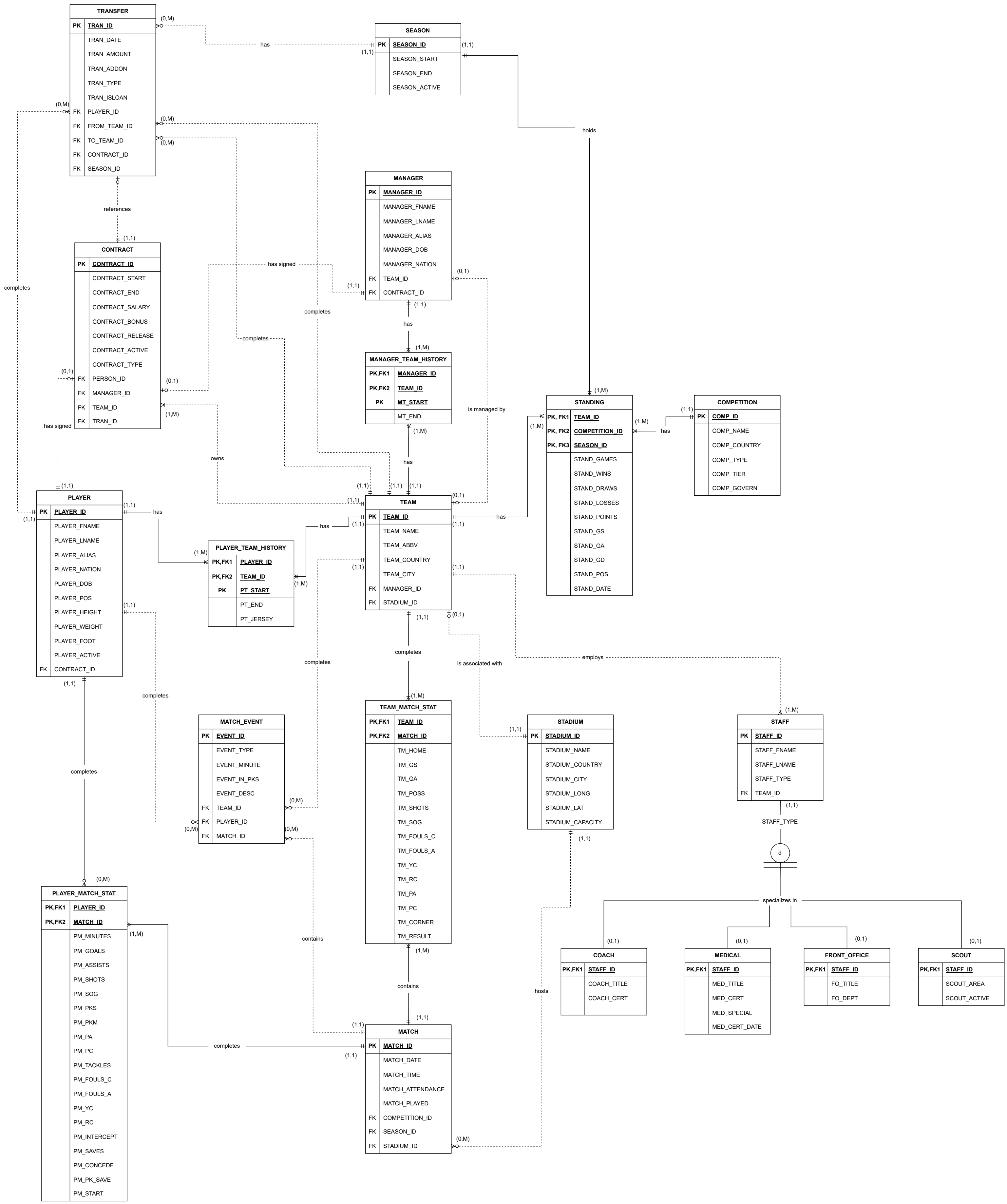
TEAM_ID	TEAM_NAME	TEAM_ABBV	TEAM_COUNTRY	TEAM_CITY	STADIUM_ID	MANAGER_ID
1	Manchester United	MU	England	Manchester	1	1
2	Aston Villa	AVL	England	Birmingham	2	2
3	Barcelona	BAR	Spain	Barcelona	3	3
4	Sao Paulo	SAP	Brazil	Sao Paulo	4	4
5	Ajax	AJX	Netherlands	Amsterdam	5	5
6	Real Betis	BET	Spain	Seville	6	6

PLAYER_TEAM_HISTORY

PLAYER_ID	TEAM_ID	PT_START	PT_END	PT_JERSEY
1	1	2016-02-25	2025-01-01	10
1	2	2025-01-01	2025-05-31	9
1	3	2025-07-01		14
2	4	2018-11-15	2020-02-23	11
2	5	2020-02-23	2022-08-30	39
2	1	2022-08-30	2025-01-25	21
2	6	2025-01-25	2025-06-30	7
2	6	2025-09-01		7

Querying the Database

1. What are the capacities of all the stadiums in England, ordered largest to smallest?
    a. SELECT STADIUM_NAME, STADIUM_CAPACITY, STADIUM_CITY, STADIUM_COUNTRY FROM STADIUM WHERE STADIUM_COUNTRY = ‘England’ ORDER BY STADIUM_CAPACITY DESC;
2. What is the current Manchester United roster?
    a. SELECT P.PLAYER_ID, P.PLAYER_FNAME, P.PLAYER_LNAME, P.PLAYER_ALIAS, P.PLAYER_POS, PTH.PT_JERSEY
    FROM PLAYER P
    JOIN PLAYER_TEAM_HISTORY PTH ON P.PLAYER_ID = PTH.PLAYER_ID
    WHERE PTH.TEAM_ID =
    (SELECT TEAM_ID FROM TEAM WHERE TEAM_NAME = ‘Manchester United’)
    AND PTH.PT_END IS NULL;
3. What is the current Premier League Table?
    a. SELECT S.STAND_POS, T.TEAM_NAME, S.STAND_POINTS, S.STAND_GAMES,
    S.STAND_WINS, S.STAND_DRAWS, S.STAND_LOSSES, S.STAND_GD,
    S.STAND_GA, S.STAND_GD
    FROM STANDING S
    JOIN TEAM T ON S.TEAM_ID = T.TEAM_ID
    JOIN COMPETITION C ON S.COMPETITION_ID = C.COMP_ID
    JOIN SEASON SE ON S.SEASON_ID = SE.SEASON_ID
    WHERE C.COMP_NAME = 'Premier League'
    AND SE.SEASON_ACTIVE = 1
    ORDER BY S.STAND_POS ASC;



TRANSFER
PK TRAN_ID
TRAN_DATE
TRAN_AMOUNT
TRAN_ADDON
TRAN_TYPE
TRAN_ISLOAN
FK PLAYER_ID
FK FROM_TEAM_ID
FK TO_TEAM_ID
FK CONTRACT_ID
FK SEASON_ID
SEASON
PK SEASON_ID
SEASON_START
SEASON_END
SEASON_ACTIVE
has
holds
references
completes
CONTRACT
PK CONTRACT_ID
CONTRACT_START
CONTRACT_END
CONTRACT_SALARY
CONTRACT_BONUS
CONTRACT_RELEASE
CONTRACT_ACTIVE
CONTRACT_TYPE
FK PERSON_ID
FK MANAGER_ID
FK TEAM_ID
FK TRAN_ID
has signed
MANAGER
PK MANAGER_ID
MANAGER_FNAME
MANAGER_LNAME
MANAGER_ALIAS
MANAGER_DOB
MANAGER_NATION
FK TEAM_ID
FK CONTRACT_ID
MANAGER_TEAM_HISTORY
PK,FK1 MANAGER_ID
PK,FK2 TEAM_ID
PK MT_START
MT_END
is managed by
owns
STANDING
PK, FK1 TEAM_ID
PK, FK2 COMPETITION_ID
PK, FK3 SEASON_ID
STAND_GAMES
STAND_WINS
STAND_DRAWS
STAND_LOSSES
STAND_POINTS
STAND_GS
STAND_GA
STAND_GD
STAND_POS
STAND_DATE
COMPETITION
PK COMP_ID
COMP_NAME
COMP_COUNTRY
COMP_TYPE
COMP_TIER
COMP_GOVERN
PLAYER
PK PLAYER_ID
PLAYER_FNAME
PLAYER_LNAME
PLAYER_ALIAS
PLAYER_NATION
PLAYER_DOB
PLAYER_POS
PLAYER_HEIGHT
PLAYER_WEIGHT
PLAYER_FOOT
PLAYER_ACTIVE
FK CONTRACT_ID
PLAYER_TEAM_HISTORY
PK,FK1 PLAYER_ID
PK,FK2 TEAM_ID
PK PT_START
PT_END
PT_JERSEY
TEAM
PK TEAM_ID
TEAM_NAME
TEAM_ABBV
TEAM_COUNTRY
TEAM_CITY
FK MANAGER_ID
FK STADIUM_ID
is associated with
employs
MATCH_EVENT
PK EVENT_ID
EVENT_TYPE
EVENT_MINUTE
EVENT_IN_PKS
EVENT_DESC
FK TEAM_ID
FK PLAYER_ID
FK MATCH_ID
TEAM_MATCH_STAT
PK,FK1 TEAM_ID
PK,FK2 MATCH_ID
TM_HOME
TM_GS
TM_GA
TM_POSS
TM_SHOTS
TM_SOG
TM_FOULS_C
TM_FOULS_A
TM_YC
TM_RC
TM_PA
TM_PC
TM_CORNER
TM_RESULT
STADIUM
PK STADIUM_ID
STADIUM_NAME
STADIUM_COUNTRY
STADIUM_CITY
STADIUM_LONG
STADIUM_LAT
STADIUM_CAPACITY
STAFF
PK STAFF_ID
STAFF_FNAME
STAFF_LNAME
STAFF_TYPE
FK TEAM_ID
STAFF_TYPE
d
specializes in
contains
hosts
PLAYER_MATCH_STAT
PK,FK1 PLAYER_ID
PK,FK2 MATCH_ID
PM_MINUTES
PM_GOALS
PM_ASSISTS
PM_SHOTS
PM_SOG
PM_PKS
PM_PKM
PM_PA
PM_PC
PM_TACKLES
PM_FOULS_C
PM_FOULS_A
PM_YC
PM_RC
PM_INTERCEPT
PM_SAVES
PM_CONCEDE
PM_PK_SAVE
PM_START
MATCH
PK MATCH_ID
MATCH_DATE
MATCH_TIME
MATCH_ATTENDANCE
MATCH_PLAYED
FK COMPETITION_ID
FK SEASON_ID
FK STADIUM_ID
COACH
PK,FK1 STAFF_ID
COACH_TITLE
COACH_CERT
MEDICAL
PK,FK1 STAFF_ID
MED_TITLE
MED_CERT
MED_SPECIAL
MED_CERT_DATE
FRONT_OFFICE
PK,FK1 STAFF_ID
FO_TITLE
FO_DEPT
SCOUT
PK,FK1 STAFF_ID
SCOUT_AREA
SCOUT_ACTIVE